

CSL510 Introduction to Systems Engineering W26

Practical 1

Discovering Systems Around Us

This is a group project. Please form a group of 5 students each.

Objective

To develop the ability to recognize, define, decompose, and analyze any real-world entity as a system.

Before you begin, discuss within your group whether the following are systems or not, and justify your answer:

- A chair
- A classroom
- A bicycle
- A smartphone
- A university
- A family
- Any AI tool
- The Indian Railway
- A village
- A tree

Please record the discussions and submit.

Part A

Find Five Systems

Student group to walk around the department/campus or in the city/villages and photograph five systems.

Not objects. Systems.

List the Stakeholders of each system

Part B1

For each selected system, answer only six questions.

Q1. What is the purpose of this system?

Q2. Who are the stakeholders?

Q3. What are the major components?

Q4. What interactions exist?

Q5. What information flows?

Q6. What would happen if one component disappeared? (Please write for at least 3 such components)

Part B2 Component (part) Analysis

Q1. Do the parts affect each other?

Q2. Do the parts together produce an effect that is different from the effect of each part on its own?

Q3. Does the effect, the behavior over time, persist in a variety of circumstances?

Q4. If one component is improved significantly, will the entire system necessarily improve? Why?

Part C

Please classify into:

Physical, Digital, Human, Social, Cyber-Physical, System-of-Systems, Adaptive Learning

Part D

Boundary Identification for all the chosen systems

Please diagrammatically present the systems.

Environment

SYSTEM

Environment

Where does the system begin?

Where does it end?

Part E

Which property of this system cannot be explained by studying any single component individually?

Part F

Concept Strategy Operation analysis of these five systems:

Concept:

- Why does this system exist?
- What human or societal problem does it solve?

Strategy:

- Why is it organized this way?
- What alternative designs could have been used?

Operations:

- How does the system function?
- How does it adapt?
- What failures may occur?

Part G

What AI can be part of these systems to augment their functionality?

Part H

Reflection on how your definition and understanding of a system changed?

Deliverables

Each student submits

- A - Five photographs, Five stakeholder lists
- B1- Five system analyses
- B2 - Component Analysis
- C - Classification
- D - Five boundary diagrams
- E
- F - CSO analysis
- G - AI augmentation
- H - Reflection
- Submission of Discussion at the beginning

Evaluation

following evaluation matrix. It emphasizes **observation, analysis, abstraction, and communication**, rather than simply correctness.

Sr. No.	Deliverable	Marks	Evaluation Criteria
A	Five Photographs & Stakeholder Lists	10	Selection of appropriate systems (5), completeness and correctness of stakeholder identification (5)
B1	Five System Analyses	15	Clarity of system purpose, components, interactions, information flow, and impact of component removal
B2	Component Analysis	10	Understanding of interdependence, emergence, persistence of behaviour, and quality of reasoning

C	Classification of Systems	5	Correct classification (physical, digital, social, cyber-physical, system-of-systems, adaptive, etc.) with justification
D	Boundary Diagrams	10	Appropriate definition of system boundary, identification of environment, clarity of representation
E	Emergence Analysis	10	Ability to identify emergent properties rather than merely listing component functions
F	Concept–Strategy–Operation (CSO) Analysis	15	Logical understanding of Concept, Strategy, and Operation with clear linkage to the chosen systems
G	AI Augmentation	10	Creativity, feasibility, relevance of AI/GenAI integration, and understanding of its role in improving system functionality
H	Reflection	10	Depth of learning, evidence of change in thinking, originality, and critical self-analysis
I	Participation in Initial Discussion	5	Active participation, quality of questions, contribution to group discussion, and originality of viewpoints
Total		100	

Please note:

1. Repeating systems amongst the groups will be checked for originality in answers.
2. Photographs should have a timestamp. (You can download the app for this)
3. Please write the group members' names and enrollment numbers on the first page.
4. Any one member should submit the deliverables.
5. Only the 'Reflection' part in the report will be submitted with the individual member's name and enrollment number at the start of each reflection.
